

## EJECTION SEATS

**THE HIGH SPEEDS ACHIEVED** through jet propulsion made a traditional parachute escape from a crippled aircraft impossible. In the forefront of the development of jets, the Germans inevitably also pioneered

ejection seats, installing them in the Heinkel He 280 prototype jet fighter from 1941. At the end of World War II the Americans lifted an ejection seat from a captured German jet and used it as the basis for developing their own.



### DUMMY SEAT

*An assistant holds up a canister containing the charge that fires an ejection seat beside a spring of the size needed to provide the same energy.*

Pilots soon found that the “bang seats” were almost as dangerous as they were necessary. At the tug of a handle, the system had to first jettison the canopy, then hurl the pilot out of the aircraft in half a second, firing him clear fast enough to miss the tail. Such rapid acceleration was bound to put enormous strain

on the airman’s body. And the aviator’s arms and legs tended to flail around during ejection, leading to some nasty amputations. No wonder that pilots in the 1950s generally preferred to try nursing a damaged jet to a landing field, even at the risk of

ending up in a fireball or a crater in the ground. Today, pilots still only activate ejection seats as a last resort. But the British Martin-Baker seats and Russian Zvezda models have become ever safer. Strapping the pilot’s arms and legs to the seat has reduced injury. Seats have also become more aerodynamically stable, and will self-right if the pilot ejects with the aircraft upside down.

### IN THE HOT SEAT

*A crewman ejects from the rear cockpit of a fighter as part of an ejector seat test. The percussive violence of the experience is obviously something to be avoided if possible.*



### DELTA-WING SOLUTION

*A Dassault Mirage III serving with the Swiss Air Force shows off its paces. The tailless delta configuration, which became virtually a Dassault trademark, was one design solution for supersonic jet fighters, combining the high-speed performance advantages of a swept-back wing with a large area for maximum lift.*

with a man in it” – had a tiny wing that was razor-thin and only 7½ft (2.3m) from base to tip.

Unprecedented power and innovative design did not spell an easy life for pilots. It was estimated that in the 1950s one in four American fighter pilots would end his career by dying in a flying accident. The F-104 was a notoriously unforgiving aircraft. When it was delivered in large numbers to West Germany, attempting to rebuild an air force disbanded after the war, it killed more than 100 pilots in a decade.

By the end of the 1950s, some test pilots in the United States and the Soviet Union were preparing for a new role as astronauts or cosmonauts. Meanwhile, the X-15, star of the Edwards Air Force Base research program from 1959 to 1968, rocketed into space – although not into orbit – reaching Mach 6.7 and an altitude of about 67 miles (108km). The heroic age of the jet pilot was transforming into the space age.



### EJECTOR SEAT

*When activated, the Martin-Baker Type 10A ejector seat shown here, fitted in the Panavia Tornado, accelerates aviators to a speed of 100mph (160kph) in 0.4 seconds.*